



Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 113: Discrete Mathematics

Batch: 26th

Date: 24/06/2024

Mid Term Examination

Semester: Spring 2024
Time: 1 hour 30 minutes

Total Marks: 30

Answer any five questions from the following.

5x6= 30

1 (a) Write short notes on Propositional Calculus and predicate Calculus.

CLO 1 2

A (b) Prove the De-Morgan's law for three bits using proposition.

CLO 1 4

2 (a) Briefly explain Universal quantifier with an example.

CLO 1 2

(b) Find the truth values for the following expression:

CLO 1 4

$[(\neg p \vee q) \wedge (\neg p \vee r)] \rightarrow ((q \wedge r) \rightarrow (p \rightarrow \neg r)) \rightarrow \neg q$, and justify the expression is either a Tautology or not.

3/4 (a) By using Mathematical Induction, prove that $\sum_{j=0}^n ar^j = \frac{ar^{n+1}-a}{r-1}$.

CLO 1 4

(b) Define Mathematical Proof techniques with its type.

CLO 4 2

4/1 (a) Define Venn Diagram with an example.

CLO 4 2

4 (b) Let U be a universal set that contains all numbers less than 20 and V be a set that contains all Prime numbers less than 13. Show the complement of V in Venn diagram. Find the power set of V and cardinality of the complement of V.

CLO 4 4

X (a) Define different types of function with diagram.

CLO 4 2

(b) Let $f_1(x) = x^3 + 2x^2 - 1$ and $f_2(x) = x^3 - 4x + 2$ be two functions. Determine the functions with reasons. Find $f_1 + f_2$ and $f_1 f_2$.

CLO 4 4

6/2 (a) Differentiate between function and relation.

CLO 4 2

2 (b) Let $A = \{1, 2, 3\}$, $B = \{1, 2, 3, 4\}$ and $C = \{0, 1, 2\}$ be 3 sets. R be a relation from A to B and S be a relation from B to C, where $R = \{(a, b) \mid a \leq b\}$ and $S = \{(b, c) \mid b \leq c\}$. Find $S \circ R$ and $R \circ S$.

CLO 4 4

7 (a) Let $A = \{1, 2, 3, 6, 24, 48\}$ be a set, find the following relations:

CLO 4 6

$R_1 = \{(a, b) \mid a \leq b\}$,

$R_2 = \{(a, b) \mid a \text{ divides } b\}$,

$R_4 = \{(a, b) \mid a^2 + b \leq 200\}$

Determine the properties with reasons of each relation. Find the matrix and graphical representation of R_1 .



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science Engineering & Technology
Department of Computer Science and Engineering

Bachelor of Science (B.Sc)

PHY 111 : Physics I

Mid-Term Examination

Spring 2024

Batch 2024

Date: 26 - June - 2024

1st Semester

Time: 1 Hour 30 Minutes

Answer the following questions

Total Marks: 30

1. Define the followings (Any 3)

CLO1 2×3=6

- a) Simple Harmonic Motion
- b) Differential equation of S.H.M
- c) Time period of simple pendulum
- d) Compound pendulum
- e) Periodic motion

2. Apply your knowledge and proof the followings (Any 3)

CLO2 5×3=15

a) Show that for a particle executing S.H.M its velocity at any point is $\frac{dy}{dt} = \omega\sqrt{a^2 - y^2}$. Where the symbols represent usual meaning.

b) Find Average kinetic energy of a vibrating particle is $\pi^2 mn^2 a^2$

Where the symbols represent usual meaning.

c) Show the time period of a simple pendulum is $T = 2\pi\sqrt{\frac{l}{g}}$. Where the symbols represent usual meaning.

d) Show the time period of a loaded spring is $T = 2\pi\sqrt{\frac{Mx}{mg}}$ Where the symbols represents usual meaning.

e) Find the time period of torsional pendulum.

3. Analyze the followings and find out the answers of the followings (Any 3)

CLO3 3×3=9

- a) For a particle vibrating simple harmonically the displacement is 8 cm at the instant the velocity is 6 cm/s and the displacement is 6 cm at the instant the velocity is 8 cm/s. Calculate (i) Amplitude (ii) Frequency

- b) A particle performs simple harmonic motion given by the equation $y = 20 \sin(\omega t + \alpha)$. If time period is 30 seconds and the particle has a displacement of 10 cm at $t=0$, find i) epoch ii) phase angle at $t=5$ seconds
- c) A particle execute simple harmonic motion given by the equation $y = 12 \sin\left(\frac{2\pi t}{10} + \frac{\pi}{4}\right)$ calculate (i) Amplitude
(ii) frequency (iii) velocity at $t=1.25$ s
- d) A spring mass of 100 gm is loaded with 200 gm of a mass. Calculate (i) the time period and (ii) and frequency if the loaded body extends 10 cm
- e) If 40 Nm torque applied to a disk and rotate the disk at 4 rad. Find time period of the disk .(Inertia of the disk is 1000 kgm^2)



CL Os	Questions	Level	Marks
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Part A

Read the following passage carefully and answer question no. 1 and 2 based on your understanding of it.

A bright mind named Ada Lovelace appeared in the wealthy salons of London at the beginning of the 1800s. She was the daughter of Lady Annabella Milbanke and poet Lord Byron.

Ada's intellectual curiosity led her to Charles Babbage, a mathematician and inventor who was designing a mechanical calculator known as the Analytical Engine. With its many gears and levers, this complex machine was intended to be a tool for performing any kind of calculation—a groundbreaking idea at the time.

Although Babbage's Analytical Engine was a complex theoretical idea, Ada was the one who realized its actual potential. She came to understand that the machine was a general-purpose computational tool as it could perform a series of operations in addition to simple math calculations.

Ada's translation and expansion of an Italian engineer's work on Babbage's machine led to the realization of this revelation. She wrote in her notes exactly how Bernoulli numbers could be computed using the Analytical Engine, so establishing the first algorithm created especially for computer application.

Her observations were imaginative, not just translations. Ada saw beyond arithmetic, forecasting the machine's ability to manipulate symbols and process any data representable by them. She also invented the idea of a "loop," where a series of instructions could be repeated. She also made predictions about how computers might be used in the arts and music, envisioning a time when machines could create intricate symphonies.

Ada Lovelace's work laid the foundation for what we now call computer programming.

Though the Analytical Engine was never built in her lifetime, her pioneering ideas lived on. Her legacy as the world's first computer programmer endures, inspiring generations of scientists and mathematicians. Ada's vision bridged the worlds of poetry and technology, forever changing the course of history with her inventive foresight.

CLO 2	1. Detect whether the statements are true or false. If false, provide the correct answer:	Understand	1x5=5
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a) Ada Lovelace predicted that machines could only be used for mathematical calculations and not for creative purposes like music or art.

- b) Ada was the daughter of the famous English poet, Lord Byron.
- c) Her notes on the Analytical Engine included the first algorithm intended to be executed by a machine.
- d) The Analytical Engine was fully built and operational during Ada Lovelace's lifetime.
- e) Ada introduced the concept of "hyper-loop"

CLO 2. Answer the following in full sentences:

Analyze 1x5=

5

- 2 a) Who is Ada Lovelace?
- b) What inspired Ada to develop an interest in mathematics and science?
- c) What is 'loop'?
- d) What is the impact of Ada's vision?
- e) How was Ada's insight on Babbage's Analytical Engine expressed?

Part B

CLO 3. Complete the text using appropriate prepositions.

Apply 0.5X

1

10=5

Student: Excuse me, when are your office hours?

Instructor: I am available (a) _____ Thursday (b) _____ 13:30 and 15:30.

Student: I have class (c) _____ that time. Can we meet (d) _____ lunch?

Instructor: Is it urgent?

Student: Yes, I have some questions to ask you (e) _____ the mid-term exam.

Instructor: You can come (f) _____ my office (g) _____ 11:00. Does that work (h) _____ you?

Student: Yes, thank you. Can you remind me where is your office?

Instructor: My office is (i) _____ the third floor (j) _____ the David Turpin Building.

Student: Thank you very much! See you then.

CLO 4. Identify the errors and replace it with the right form of verbs.

Apply 0.5X

1

10=5

Nowadays, the news of Artificial intelligence (AI) are hot topic. AI have transforming many aspects of our lives. It helping in automating tasks that was previously done by humans, making processes more efficient. AI technologies, such as machine learning and neural networks, enables computers to learn from data and make decisions. They have find applications in diverse : fields, including healthcare, finance, and transportation. For instance, AI systems can predicts disease outbreaks and assists doctors in diagnosing illnesses. In finance, AI-powered algorithms manages investments and detects fraud. Autonomous vehicles, which is a product of AI, are being developed to reduce human errors on the road. As AI continues to evolve, it create new opportunities and challenges, requiring society to adapt and develops appropriate regulations.

Part C

CLO 5. Suppose you are a tech-savvy. Now, write a 300-words paragraph on "Computer".

Apply 1x10

3

=10

Do not forget to generate a perfect title for your paragraph.



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science Engineering & Technology
Department of Computer Science and Engineering
Bachelor of Science (B.Sc)
EEE 117: Electrical Engineering
Mid-Term Examination
Spring 2024

Batch 2024

Date: 3-July-2024

1st Semester

Time: 1 Hour 30 Minutes

Answer the following questions

Total Marks: 30

1. Define the followings (Any 6)

CLO1 1×6=6

- I. Electrical circuit
- II. Electrical current
- III. Electrical voltage
- IV. Dependent and Independent source
- V. Ohm's Law
- VI. Kirchhoff's Current Law
- VII. Kirchhoff's Voltage Law

2. Apply your knowledge and answer the followings (Any 3)

CLO2 4×3=12

a) In a household, a 120W personal computer (PC) is un for 4 h/day, while a 60W bulb runs for 8 h/day. If the utility company charges Tk. 5.50 /kWh, calculate how much the household pays per year on the PC and the bulb.

b) A 12-V car battery supported a current of 150 mA to a bulb. Calculate:

(i) the power absorbed by the bulb, (ii) the energy absorbed by the bulb over an interval of 20 minutes.

c) For the circuit shown in Fig. 2 (c) calculate the voltage v and the power p .

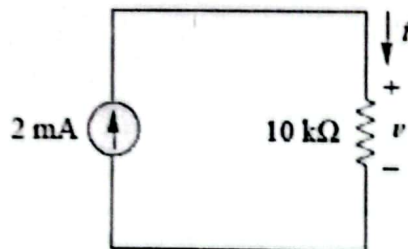


Fig. 2(c)

d) Transform the wye network in Fig. 2(d) to a delta network.

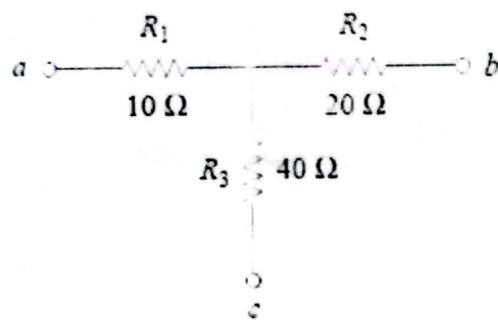


Fig. 2(d)

3. Analyze the followings and find out the answers of the followings (Any 2)

CLO3 $6 \times 2 = 12$

a) Find v_o and i_o in the circuit of Fig. 3(a).

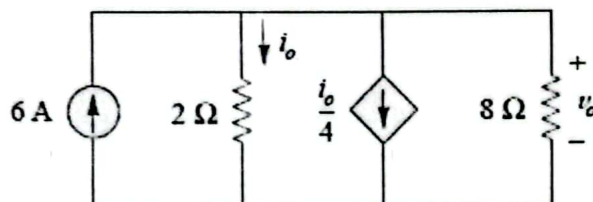


Fig. 3(a)

b) Determine the number of branches and nodes in the circuit shown in Fig. 3

(b). Identify which elements are in series and which are in parallel.

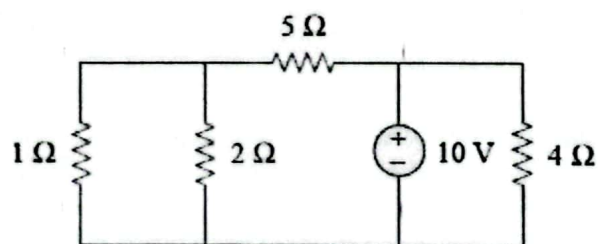


Fig. 3(b)

c) Find i and v_o from Fig. 3(c)

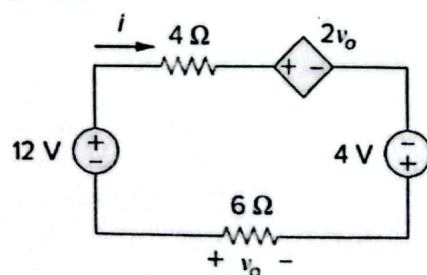


Fig. 3(c)



HAMDARD UNIVERSITY BANGLADESH

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Faculty of Science, Engineering and Technology

1st Year 1st Semester Examination 2024

Course Code: MAT115

Course Title: Mathematics-I (Differential and Integral Calculus)

Mid-Term Examination

Semester: Spring 2024

Date: 6/29/2024

Total Time: 1HR: 30 Mins.

Total Marks: 30

Answer any Three (03) out of the following questions:

3 × 10 = 30

[The figures in the right margin indicate full marks. Symbols used have their usual meaning.]

- | | | CLOs | Marks |
|-----|--|---------------|-------|
| Q1. | (a) Define function. Justify whether the followings are functions or not.
(i) $y = \sqrt{x^3}$, (ii) $y^2 = x$, (iii) $ y = x$. | CLO1,
CLO2 | [4] |
| | (b) What is a periodic function? Show that the function $f(x) = \tan x$ is a function of period π and the function $g(x) = \cos x$ is a function of period 2π . | CLO1,
CLO2 | [2] |
| | (c) Check for the existence of limit at $x = 1$ and $x = 2$ for the following function
$f(x) = \begin{cases} x^2 & \text{when } x > 2 \\ 2.5 & \text{when } 1 \leq x \leq 2 \\ x^2 + 2 & \text{when } x < 1 \end{cases}$ | CLO3 | [4] |
| Q2. | (a) Define continuity of a function at a point. Also distinguish between $f(a)$ and $\lim_{x \rightarrow a} f(x)$. | CLO1 | [2] |
| | (b) A function $f(x)$ is defined by
$f(x) = \begin{cases} 3 + 2x & \text{for } -\frac{3}{2} \leq x < 0 \\ 3 - 2x & \text{for } 0 \leq x < \frac{3}{2} \\ -3 - 2x & \text{for } x \geq \frac{3}{2} \end{cases}$ | CLO3 | [5] |
| | Check the continuity of $f(x)$ at $x = 0$ and $x = \frac{3}{2}$.
Determine the limits of the followings: | | |
| | (c) (i) $\lim_{x \rightarrow 0} \frac{1 - \cos 7x}{7x^2}$ | CLO3 | [3] |
| | (ii) $\lim_{x \rightarrow \infty} \{\ln(x^3 - 8) - \ln(x^2 - 4)\}$. | | |
| Q3. | (a) When a function is differentiable at a point? A function $f(x)$ is defined as
$f(x) = \begin{cases} 1 + 2x & \text{for } -\frac{1}{2} \leq x < 0 \\ 1 - 2x & \text{for } 0 \leq x \leq \frac{1}{2} \\ -1 + 2x & \text{for } x \geq \frac{1}{2} \end{cases}$ | CLO1,
CLO3 | [4] |
| | Check whether the function $f(x)$ is differentiable at $x = 0$ or not. | | |
| | (b) Determine $\frac{dy}{dx}$ for the followings:
(i) $x^y = y^x$ | CLO3 | [6] |
| | (ii) $y = (\cos x)^{\sin x} + (\sin x)^{\cos x}$ | | |
| Q4. | (a) State Leibnitz theorem. If $y = \cos(m \cos^{-1} x)$, then show that
(i) $(1 - x^2)y_2 - xy_1 + m^2y = 0$ | CLO2,
CLO3 | [6] |
| | (ii) $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - (n^2 - m^2)y_n = 0$ | | |



- (b) What do you mean by maxima and minima of a function? If

CLO2 [4]

$f(x) = x + \frac{1}{x}$, then show that the minimum value of the function is greater than the maximum value.

- Q5. (a) Define partial differentiation of a function of two variables.

CLO1, CLO3 [5]

If $f(x) = \ln(x^3 + y^3 + z^3 - 3xyz)$, then show that

(i)
$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} + \frac{\partial f}{\partial z} = \frac{3}{x+y+z}.$$

(ii)
$$\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right)^2 f = -\frac{9}{(x+y+z)^2}$$

- (b) Define homogeneous function of three variables. State Euler's theorem for homogeneous functions and verify it for the function

CLO2, CLO3 [5]

$$f(x, y, z) = x^2 + y^2 + z^2 - xy - yz - zx$$



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 117 : Engineering Ethics and Cyber Law

Batch: 26th

Date: 06/07/2024

Midterm Examination

Semester: Spring 2024
Time: 1 Hour 30 Minutes
Total Marks: 30

Answer any five questions.

5 × 6 = 30

- | | |
|---|----------------------------------|
| 1. (a) Define any two the following terms:
i. Ethics
ii. Moral
iii. Utilitarianism
(b) Explain the significant skills for Ethical Reasoning. | CLO2 2

CLO2 4 |
| 2. (a) An engineer who develops a proto-type for the project, loses it due to a mishap exactly the day before the submission. Is it morally correct to outsource the prototype of the project and reduce the risks of job insecurity? What types of inquiries he should do? | CLO5 6 |
| 3. (a) What is Moral Dilemmas?
(b) Briefly explain the types of complexities of Moral Dilemmas. | CLO2 2
CLO2 4 |
| 4. (a) Write the full meaning of the given abbreviations-
i. NSPE
ii. ASCE
iii. ASME
iv. BCC
(b) Write the fundamental of principles of code of ethics of engineers. | CLO4 2

CLO2 4 |
| 5. Explain the Lawrence Kohlberg's Theory on moral development. | CLO1 6 |
| 6. (a) Define Moral Autonomy.
(b) Explain the skills needed for Moral Autonomy. | CLO2 1
CLO2 5 |
| 7. (a) What is profession? Write the main criteria of a profession.
(b) Explain the models of professional engineers. | CLO2 3
CLO2 3 |